



Predicting the Employment Rate of Students After Graduation Using Machine Learning Methods

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Abstract. In the context of the labor market is volatile, requires candidates to have the ability and skills to match the needs of the employers. However, most recent graduates have difficulty finding jobs that match their abilities. At the same time, assessing the ability of candidates and selecting the right candidates for the job is a huge challenge for the employers. In this paper, we introduce a new approach to early prediction the employable of students in Vietnam after graduation by using some machine learning methods: Gaussian Naive Bayes (NB), Support Vector Machine (SVM), Decision Tree (DT), Random Forest (RF), and Artificial Neural Network (ANN). We collect data about students from the graduation year of 2020 and 2021, including learning scores, working environment, and personal information such as gender, hometown. After that, we create three models to early predict the students employment status from learning data of first academic year (Model_1), after the first two years of study (Model_2) and fourth year (Model_full). Beside that, three more models have been created from learning data about the subjects of ideological and political (Model_3), pedagogical skills (Model_4), foreign language skills (Model_5). Furthermore, we trying to improve the performance of the models through sampling methods such as: Synthetic Minority Oversampling Technique (SMOTE – Over-sampling), Edited Nearest Neighbors (ENN – Under-sampling) and SMOTE_ENN (Hybrid-sampling). Experimental results show that ANN methods bring the best performance to predict student employment status with accuracy up to 80% for the test dataset. Moreover, Model_1 brings positive results in terms of early prediction of student's employment status after the first year of study with an accuracy for about 70% on the test dataset.

Keywords: Job Vacancy · Job Prediction · Data Mining · Machine Learning

1 Introduction

The situation of graduates unable to find jobs has been a dilemma for many years now. Analysis report on Vietnam's education sector for the period 2011–2020 made by the Vietnam Institute of Educational Sciences in collaboration with UNESCO in Vietnam,

citing data from the General Statistics Office, in 2019, more than 3% of college graduates and 2.8% of university graduates were unemployed, while only 1.1% of intermediate education graduates were unemployed. Ho Chi Minh City, in 2022, there are nearly 150,000 unemployed workers receiving unemployment benefits include 83,000 unskilled workers and nearly 46,000 workers with university degrees or higher. Thus, up to 88% of unemployed workers are unskilled and university graduates. According to survey data about nearly 2,000 students from the graduation year of 2020 and 2021 after 1 year of graduation conducted by the Center for Quality Assurance of Hanoi National University of Education, there are 10% of students who still can't found a job, in that case 40% of students keep learning to higher levels and 60% of students actually couldn't find a job. The remainder is a group of students working in government and private enterprise environments with 40% and 50% of the total students surveyed, respectively. Hence, if it is possible to apply machine learning and data mining methods to exploit important information in student survey data, it can bring a lot of benefits to the University in making early decisions to overcome unemployment among students after graduation as well as lecturers can actively adjust the teaching content so that after graduation, students can meet the requirements of employers.

2 Related Works

Data mining algorithms have been applied in education and demonstrated through many studies. The predicting model about career field that engineering students will choose after graduation is made by Akanksha Pandey et al. [1] by using popular classification algorithms such as K-nearest neighbors, SVM, Logistic regression, Decision trees, Gaussian Naive Bayes and Artificial Neural Networks. The study shows that K-nearest neighbors algorithm gives the best results with an accuracy of 63.4% while the artificial neural network model has the lowest accuracy with 45.45%.

In recent years, machine learning and deep learning techniques are tending to be used in the field of big data mining in education as well as in labor market analysis [2–11]. Teng Guo et al. [8] have researched and built a MAYA (Multi-mAyor emploYment stAtus) model to predict the employment status of students after graduation with bias. The results show that the models based on popular machine learning methods give stable results with the original dataset, but after the authors used GAN to solve the data imbalance problem, the MAYA model shows outstanding results with up to 88% accuracy.

Fen Yang's research group [9] at Henan University, China built a model to predict the career trends of students after graduation by using decision tree algorithms based on information fields such as areas of expertise, learning outcomes at university, foreign language skills and computer skills. Research results show that foreign language skills and computer skills have the strongest influence on student's career trends.

Xing Li and Ting Yang [10] used Long short-term memory (LSTM) network to build a forecasting model of employment situation of Chinese graduates based on data statistical method with cluster analysis technology model. Through the research results, it can be seen that the employment rate of Chinese university graduates tends to be relatively stable, the possibility of an increase or decrease of this rate is relatively small.

A new approach based on student's personal data and academic performance has been developed by Yan Wei and colleagues [11] to predict students' entrepreneurial intentions. The prediction model is mainly based on the kernel extreme learning machine (KELM) optimized by the improved Harris hawk's optimizer (HHO). In order to obtain better parameters and feature subsets, the Gaussian barebone (GB) strategy is introduced to improve the HHO algorithm, so as to strengthen the optimization ability for tuning parameters of KELM and identifying the compact feature subsets. Then, an optimal KELM model (GBHHO-KELM) is established according to the obtained optimal parameters and feature subsets to predict the entrepreneurial intention of students. Compared with other machine learning algorithms tested by the authors, GBHHO-KELM has the highest prediction accuracy (90%) and more stable performance in predicting student's entrepreneurial intentions.

3 Experiment

3.1 Proposed Methods

In this study, we propose several methods to build models predicting student employment status from their learning data by combining resampling techniques and selected machine learning-based classifiers. Three different resampling techniques were applied to balance the dataset including Under-sampling - Edited Nearest Neighbors (ENN), Over-sampling - Synthetic Minority Oversampling Technique (SMOTE) and Hybrid sampling - SMOTE_ENN. Besides, the original dataset is still kept to build models and compare performance with the other resampling techniques. To have the most intuitive view of the ability of machine learning methods to predict student's employment status, we use artificial neural network and compare with popular machine learning-based classifiers such as Gaussian Naive Bayes (NB), Support Vector Machine (SVM), Decision Tree (DT), Random Forest (RF) to find the best method via the confusion matrix, accuracy, precision, recall, Cohen's Kappa and F1-score. Figure 1 describe the research implementation process.

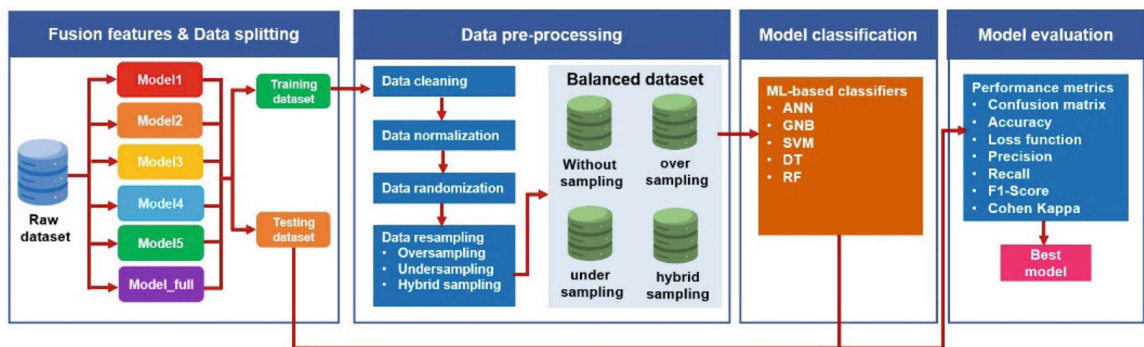


Fig. 1. Proposed research method.

3.2 Data Collection

The dataset in this research mainly about students from the graduation year of 2020 and 2021, including learning scores, working environment, and personal information such as gender, hometown. Surrounded by these graduates, 1567 students were surveyed after one year graduate as research objects, of which 800 students were working in private enterprise, 631 students were fully employed by government and 136 students are jobless. We collecting 19 information fields about student's personal data such as gender, hometown, working environment and student's university score for example foreign language, specialized foreign language, ideology and politics, pedagogical skills, internship and employment status. The goal is trying to explore the potential importance and the internal correlation of these attributes as well as to build a machine learning model for predicting the employment rate of students after graduation.

3.3 Data Analysis

The issue that the society as well as the university most concerned about is the working environment and employment situation of students after graduation, as presented in Fig. 2. The working environment of alumni is very diverse, but it also gives a good signal when the students with the right major working environment account for the most about 72.8%.

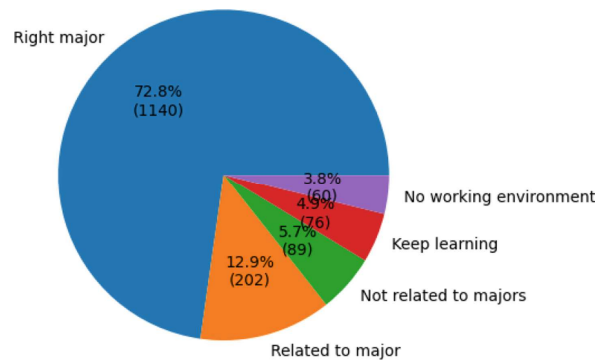


Fig. 2. Student's statistics by working environment.

Basically, the employment status of students is divided into three main groups: private enterprises, government and jobless. To get an overview of the employment status of students after graduation, we summarize the employment and unemployment rates of students by graduation grade as shown in Fig. 3.

It can be seen that students graduating with an very good, good and average degree tend to work in private enterprises. At the excellent degree, the number of students working in the government is 49%, slightly higher than the number working in private enterprises with 42%, this group has a low unemployment rate of 9%.

Last but not least, we create a graph of the correlation between the information fields in the collected dataset, as shown in Fig. 4. The correlation coefficient that measures the strength and weakness of the relationship between two attributes in the dataset is shown by the heatmap below.

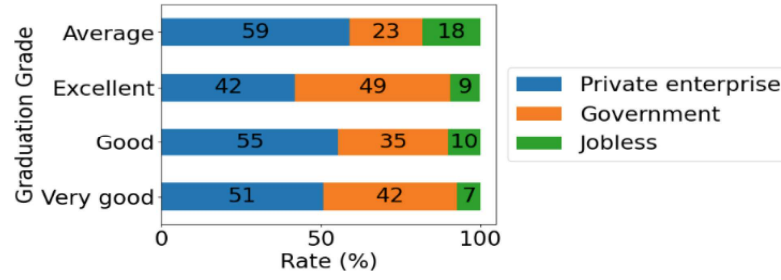


Fig. 3. Employment status according to graduation rating after 1 year of graduation.

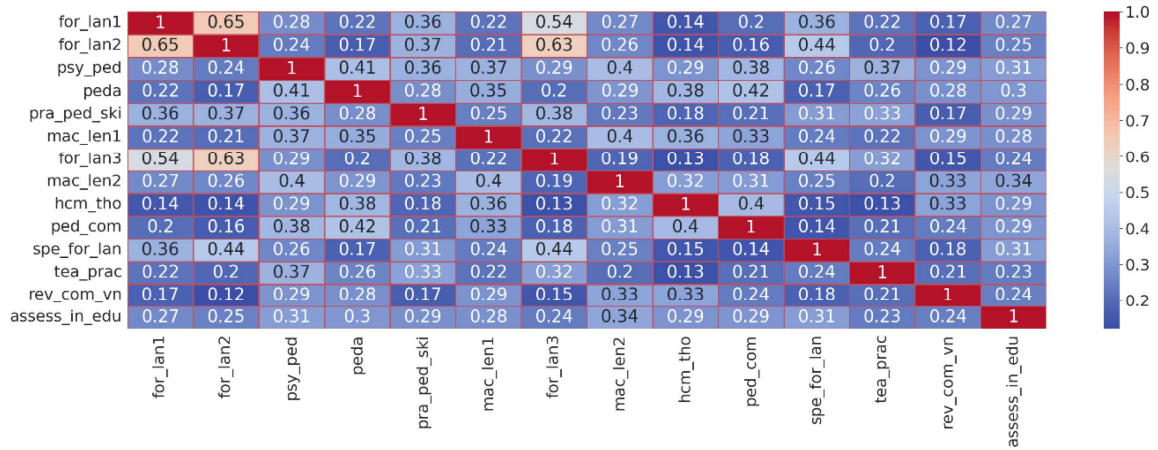


Fig. 4. The correlation coefficient between data fields.

It can be seen that subjects in the same field are highly correlated. For example, subjects in foreign language skills have a correlation coefficient of about 0.53–0.65, while subjects in the field of education have a correlation coefficient of about 0.4. The same thing happened with the group of subjects on political thought. For those reasons, we decided to create a number of different predictive models based on the input fields of scores from subjects in the same field.

3.4 Data Preprocessing

The data preprocessing process follows the usual steps in the field of machine learning. To fit the dataset for the input of predictive models, several feature engineering methods are used such as ordinal encoder, label encoder and one-hot encoder. To be able to improve the performance of the predicting models, we use SMOTE technique [12], ENN [13] and SMOTENN [14], to solve the problem of imbalance in the dataset. Synthetic Minority Oversampling Technique (SMOTE) is a statistical technique for increasing the number of cases in the dataset by a balanced way. Therefore, after applying SMOTE, the number of data samples of all classes are equal to the majority class private enterprise which is 800 and the total number of samples in the dataset increases to 2400.

3.5 Create Predicting Model

To predict the employment situation of students, we create six models in which three models to early predict the students employment status from learning data of first

academic year (Model_1), after the first two years of study (Model_2), fourth year (Model_full) and three more models have been created by field of subjects about thought and politics (Model_3), pedagogical skills (Model_4), foreign language skills (Model_5). Figure 5 shows how the features are used for the respective models. To evaluate the performance of models in predicting student employment status, this study uses machine learning-based methods such as Gaussian Naive Bayes (NB), Support Vector Machine (SVM), Decision Tree (DT), Random Forest (RF), and Artificial Neural Network (ANN).

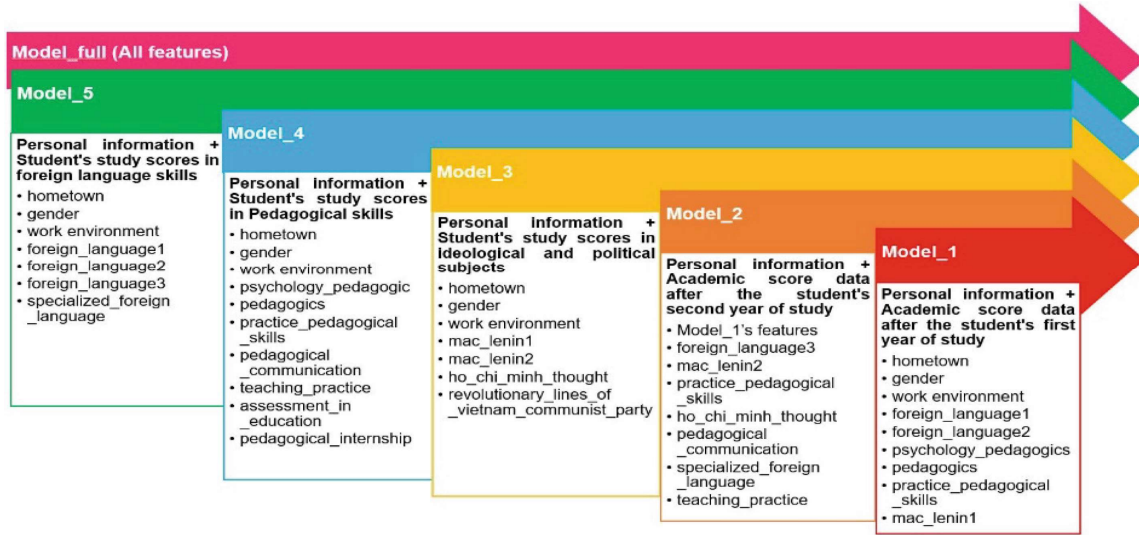


Fig. 5. Features used to train prediction models.

4 Results

4.1 Training Models

After experimenting with different parameters with the ANN model, we decided to train the model with 100 epochs and batch size of 16 to get stable performance. The results show that training loss and validation loss tends to decrease after each epoch in all models. In contrast, training accuracy and validation accuracy tend to increase after each training epoch.

Other models are also trained and tested on the same dataset as the ANN model to give the most objective results. In this study, NB model uses the var_smoothing parameter with a value of $1e-9$ to help the model not be too sensitive to outliers. SVM model uses the default regularization parameter C equal to 1.0 with squared L2 penalty, radial basis function kernel and gamma index aligned in the ratio $1/(n_features * X.var())$ with X being the training data sample and var() being the covariance calculation method. DT model with the strategy to divide at each node is “best strategy”, the function to evaluate the quality of division is Gini impurity. RF model initializes 100 trees in the forest, the partition quality evaluation function is still Gini impurity.

4.2 Confusion Matrix Evaluate

The performance of ANN models is even more clearly demonstrated when evaluated with the test dataset, as presented in Fig. 6. ANN models that do not use resampling techniques have the highest error rates, mainly coming from government and private enterprise data samples. In terms of the ability to predict data samples belonging to the government class, Model_4 using original dataset performed best with recall about 1.0. With data samples belonging to the private enterprise class, Model_4 using SMOTE resampling bring best results.

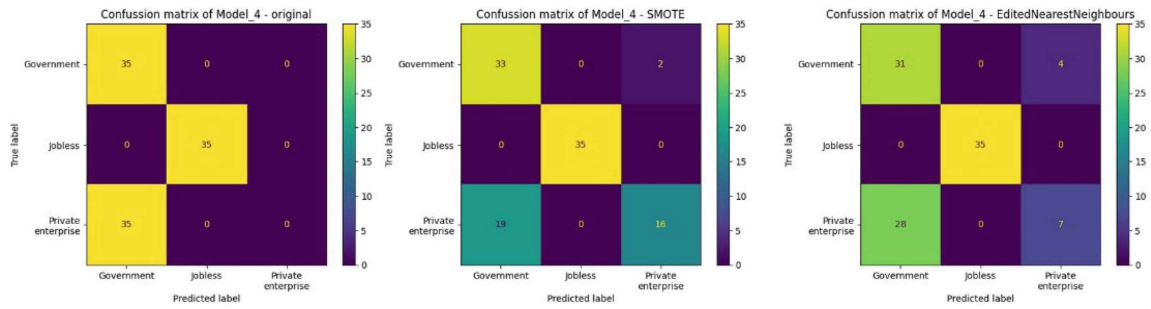


Fig. 6. Confusion matrix of best models without resampling.

All ANN models provide absolute accuracy when predicting data samples of the jobless class. Turns out, through the test dataset, it can be seen that the errors of the models do not come from the jobless class but from the government and private enterprise classes.

4.3 Analyze the Influence of Features

To explore the factors that influence employability in the dataset, we use the SHAP (SHapley Additive Explanations) technique, one of the popular techniques in the field of Explainable AI (XAI). SHAP [15] is a unified framework for explaining the predictions of complex machine learning models, helping to determine the importance of each feature to a particular prediction. SHAP uses Shapley value theory from game theory to calculate the contribution of each feature to the model's prediction. The analysis results are shown in Fig. 7.

It can be seen that people with suitable majors, long-term study plans, and foreign language proficiency will have a high chance of working in government agencies. In addition, lack of work environment or lack of relevance between job and major will increase the possibility of unemployment. With private companies, there is more flexibility in choosing jobs, sometimes students doing a job that is not right or related to their major. People working in government agencies and private companies tend to continue their education at higher levels. For students who are unemployed after graduation, they are all affected by continuing their postgraduate studies or are still looking for a job in right major.

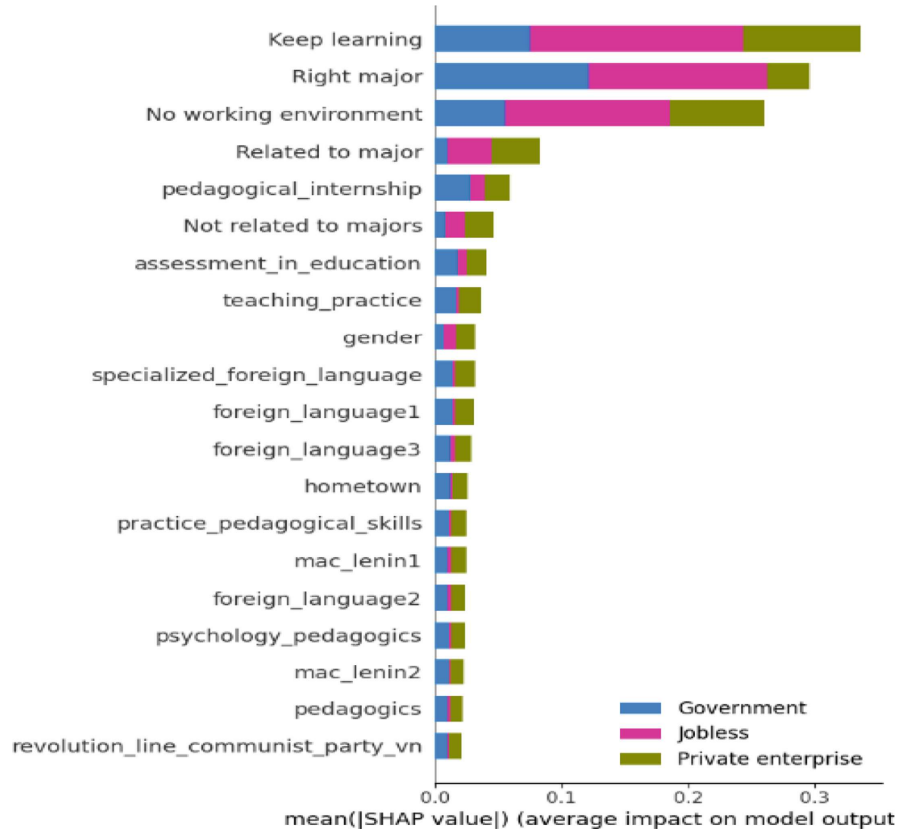


Fig. 7. SHAP value average impact on model output.

4.4 Evaluate Prediction Models

The prediction results of the models after applying different machine learning methods and resampling techniques are shown in Table 1. When compared with other machine learning methods, ANN models are not too superior, but show their strengths in some specific cases. With the test dataset without using any resampling technique, ANN maintains stability with an accuracy of 67%, other methods are slightly higher accuracy about 70%. For Model_5, the DT method gives the highest accuracy of 71%.

Table 1. Compare the accuracy of all models when applying different machine learning methods and resampling techniques on the test dataset.

Test set	Without resampling					SMOTE					ENN					SMOTE_ENN				
	ANN	NB	SVM	DT	RF	ANN	NB	SV M	DT	RF	ANN	NB	SV M	DT	RF	ANN	NB	SV M	DT	RF
Model_1	0.67	0.7	0.7	0.63	0.7	0.7	0.7	0.69	0.67	0.65	0.66	0.7	0.67	0.66	0.65	0.67	0.69	0.67	0.67	0.68
Model_2	0.67	0.7	0.7	0.67	0.7	0.67	0.69	0.66	0.71	0.68	0.64	0.7	0.69	0.66	0.66	0.65	0.7	0.7	0.7	0.71
Model_3	0.67	0.7	0.7	0.64	0.7	0.68	0.7	0.68	0.7	0.7	0.67	0.7	0.67	0.7	0.66	0.65	0.7	0.73	0.66	0.7
Model_4	0.67	0.7	0.7	0.66	0.7	0.8	0.71	0.73	0.77	0.71	0.7	0.7	0.77	0.72	0.76	0.67	0.70	0.71	0.76	0.7
Model_5	0.67	0.7	0.7	0.71	0.7	0.66	0.7	0.7	0.65	0.74	0.68	0.7	0.65	0.68	0.67	0.67	0.7	0.7	0.66	0.7
Model_full	0.67	0.7	0.7	0.7	0.69	0.73	0.7	0.7	0.68	0.69	0.66	0.7	0.68	0.68	0.65	0.67	0.71	0.77	0.74	0.75

When using ENN resampling technique, the ANN model did not provide better accuracy than other machine learning models. The prediction model based on data from information fields related to pedagogical skills (Model_4) using SVM has the highest accuracy of up to 77%. ANN models has shown its effectiveness in predicting the test dataset using SMOTE resampling technique. Model_4 using ANN has given the highest accuracy up to 80%. The SMOTE_ENN resampling technique helps increase the number of data samples in the jobless class and reduce the number of data samples in other classes. Meanwhile, jobless data samples are inherently predicted by the models with almost absolute accuracy, this makes the accuracy of the models when predicting the test dataset doesn't change too much. The highest accuracy is 77% belongs to Model_full using SVM method and SMOTE_ENN.

5 Discussion

From the experimental results of this research, it is easy to see that the percentage of students working in their respective majors accounts for the majority. In which, the very good, good and average graduates tend to work for private enterprises more, while the excellent graduates prefer to work in government agencies. Besides, based on the correlation coefficient chart, subjects in the same field will have a closer relationship, especially students who are good at pedagogical skills. It can be seen through the prediction results, Model_4 always gives good results compared to the common ground.

Through comparing the performance of machine learning methods, we know that the ANN model has a very good learning ability, but when it comes to new datasets, the results are slightly weaker than other machine learning methods such as NB, SVM and RF. The reason could be that the ANN model needs a larger number of training data samples, ranging from 10000 to 30000 samples. While other machine learning methods have always remained stable despite the small amount of data samples.

6 Conclusion and Future Work

In this article, we have analyzed and given statistics on the learning and employment status of students at Hanoi National University of Education. At the same time, an approach in building models to predict the working status of students after one year of graduation was introduced. The results show that without using any resampling techniques, the models predicting early employment status from first year student and second year student data have an accuracy of 70% when predicting the test dataset. The best model without resampling is Model_5 using DT on the test dataset with prediction accuracy of 71%. The ENN resampling method yields up to 77% accuracy on the test dataset when applying Model_4 along with the SVM classifier. For SMOTE resampling, the ANN classifier wins on the test dataset when combined with Model_4 giving an accuracy of 80%. Finally, SMOTE_ENN resampling brings an accuracy on the test dataset is 77% for Model_full combined with SVM, and 76% for Model_4 with DT.

In the future, the research team will try to collect more data as well as appropriate methods to improve the performance of the predictive model. Especially researching and implementing the method of model fusion to increase accuracy.

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